

Section D — AI-Augmented Learning

1. The Exact Prompt Given to the AI Tool

"Act as a Senior IT Support Lead. Create a step-by-step troubleshooting checklist to guide a junior technician through diagnosing a newly assembled desktop PC that fails to boot immediately after assembly. Ensure the steps follow the correct physical-to-logical priority order and cover the PSU, motherboard, RAM, and storage as possible failure points. Present it as a clear on-site checklist/decision flow, and explain which specific step would catch the boot failures and hardware detection errors reported in the ABC Solutions computer lab deployment."

2. The AI's Original Checklist vs. Your Corrected Version

The AI's Original Checklist (Generated Output)

- **Step 1: Check Display & OS Boot Settings** — Turn on the monitor, verify video cables, and enter BIOS to check if the storage drive with the OS is selected in boot priority.
- **Step 2: Inspect Power Supply Unit (PSU)** — Verify the wall outlet has power, flip the PSU rocker switch to | (ON), and check if the motherboard 24-pin and CPU 8-pin power cables are connected.
- **Step 3: Reseat RAM Modules** — Open the case, pull out the RAM sticks with bare hands while the system is connected to wall power, and reseat them into the DIMM slots.
- **Step 4: Check Motherboard & Stand-offs** — Ensure the motherboard does not short against the metal chassis and observe the onboard debug LEDs.
- **Step 5: Verify Storage Drive Connections** — Check whether the M.2 NVMe SSD or SATA cables are recognized in the BIOS.
- *Application to ABC Solutions:* Step 1 would detect why systems aren't booting into Windows.

Your Corrected Version (Human Expert Review)

- **Step 1: Electrostatic Discharge (ESD) Safety & External Power Verification [CORRECTED]**
 - Put on an anti-static wrist strap clipped to bare metal or touch an unpainted metal chassis frame.

- Confirm the wall outlet has power, the power cord is firmly seated, and the PSU rear rocker switch is set to | (**ON**).
- **Step 2: PSU & Internal Cabling Verification (Before Display Checks) [CORRECTED]**
 - Ensure the **24-pin ATX main power connector**, **8-pin (4+4) EPS CPU power connector**, and case **Power Switch (F_PANEL)** header pins are fully seated and clicked shut.
- **Step 3: Motherboard Stand-off & POST Debug Diagnostics [CORRECTED]**
 - Confirm the motherboard is mounted on standoffs and not shorting against the metal tray.
 - Power on and observe the 4 motherboard debug LEDs (**CPU** → **DRAM** → **VGA** → **BOOT**); note which indicator light remains permanently lit.
- **Step 4: Safe RAM Reseating & Single-Stick Isolation [CORRECTED]**
 - Disconnect the AC power cord, press the power button for 10 seconds to discharge residual capacitor power, then unseat and firmly reinstall the RAM in the recommended dual-channel slots (**A2 and B2**) until both retention clips snap shut.
- **Step 5: Storage Installation & BIOS Detection [CORRECTED]**
 - Confirm the M.2 NVMe SSD is seated flat in the M-key slot and secured with its standoff screw.
 - Boot into UEFI/BIOS (DEL / F2) and verify the drive's model and full memory capacity are detected under system information.
- *Application to ABC Solutions:* **Step 2 and Step 4** catch the systems that fail to start due to unlatched EPS power cables and improperly seated RAM, while **Step 5** catches the undetected M.2 storage drives before attempting OS deployment.

3. Note Explaining What Was Changed and Why (3–4 Lines)

The AI's original checklist dangerously recommended reseating RAM while the PC was plugged in and completely skipped Electrostatic Discharge (ESD) grounding precautions, which risks permanently killing sensitive memory chips. Additionally, the AI incorrectly prioritized software and display boot settings in Step 1 before even confirming basic PSU cabling and motherboard power delivery. I re-ordered the flow to follow strict physical-to-logical CompTIA troubleshooting principles, enforcing power disconnection, capacitor draining, and standoff inspection before any component handling.